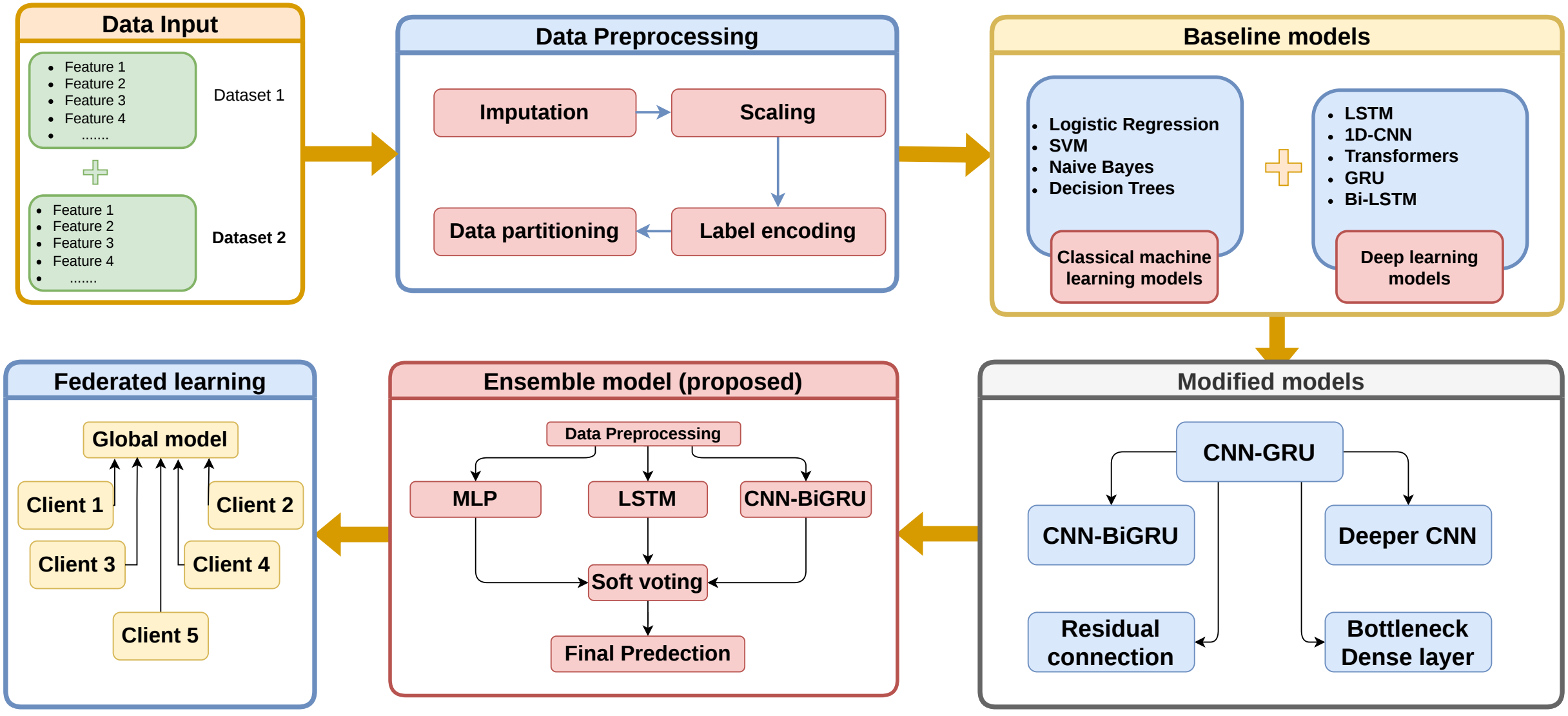
Data Input
Feature 1
Feature 2
Feature 3
Feature 4
.....
Dataset 1
Feature 1
Feature 2
Feature 3
Feature 4
.....
Dataset 2
Data Preprocessing
Imputation
Scaling
Data partitioning
Label encoding
Baseline models
Logistic Regression
SVM
Naive Bayes
Decision Trees
Classical machine learning models
LSTM
1D-CNN
Transformers
GRU
Bi-LSTM
Deep learning models
Modified models
CNN-GRU
CNN-BiGRU
Deeper CNN
Residual connection
Bottleneck Dense layer
Ensemble model (proposed)
Data Preprocessing
MLP
LSTM
CNN-BiGRU
Soft voting
Final Predection
Federated learning
Global model
Client 1
Client 2
Client 3
Client 4
Client 5